EE-312L Signals and Systems Lab
Laboratory Session 01
Introduction of MATLAB Environment and Running Basic MATLAB Commands

October 3, 2024

Outlines

1. Objective
2. Introduction to MATLAB environment
3. Variables, constants and expression
4. MATLAB Flow Control
5. MATRICES
6. MATLAB Files
7. User-defined Functions in MATLAB

1 Objective

The objective of this lab session is to introduce MATLAB and MATLAB working environment and apply the basic commands.

2 Introduction to MATLAB

MATLAB stands for MATrix LABoratory and the software is built up around vectors and matrices. This lab course is used to allow students to become familiar and work with MATLAB to solve applications related to signals and systems. After starting MATLAB there will be a number of windows you can find as shown in Figure 1:

- The Command Window

- The Command History
- The Workspace
- The Current Directory
- The Help Browser
- Creating and Saving a Matlab File: MyFile.m

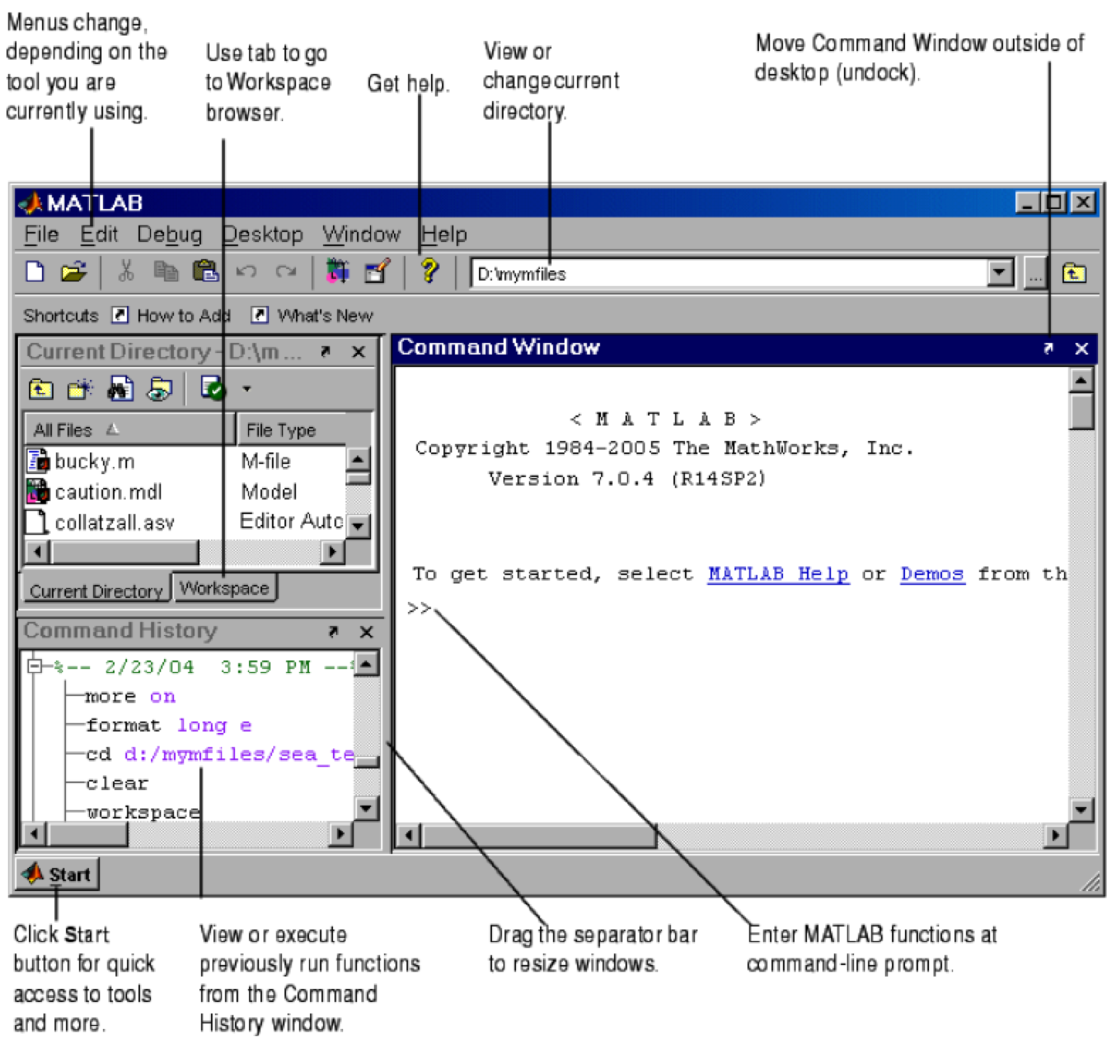


Figure 1: Graphical user interface of Matlab [?]

MATLAB can be used as a calculator by simply writing as:

```
>> 1+3
  ans=
    4
```

3 Variables, Constants and Expressions

```
variable_name = constant_value or expression
a=9;
b=2*4;
c=3*a+c;
```

A variable name may contain any character from alphabets (a-z or A-Z), integers (0-9) or special character underscore (_). Further a variable must be start with an alphabet. Matlab is case sensitive and treats A and a as two different variable names. A variable may be assigned a constant or expression. The variable name is always written on left hand side (LHS) and the value or expression is kept on right hand side (RHS). The assignment operator = is used assign an expression or value to a variable.

Matlab constant can be scalar, vector or matrix in nature with any numerical values.

```
Scalar: 3;
Vector: 1:1:5;  or [1 2 3 4 5];
Matrix: [1 2 3; 4 5 6; 7 8 9] ;
```

3.1 Built in Functions in Matlab and Help

In Matlab there are built in functions for mathematical, algebra, geometry, graphics etc.

- Elementary Math: sqr(), sqrt(), ceil(), floor(), round()
- Trigonometric: cos(), sin()
- Exponential: exp()
- Graphical: plot(), bar(), stem(),
- Coordinate System: car2pol(), pol2cart()
- Special Matrices: eye(), diag(),rand()
- Help: help cos, help sin, help plot

3.2 MATLAB Relational Operators

Less than	$<$
Less than or equal to	$\leq$
Greater than	$>$
Greater than or equal to	$\geq$
Equal to	$==$
No equal to	$\sim=$

4 Flow Control

- if statement
- switch statement
- for loop
- while loop
- break statement

4.1 if statement

```
if expression
 statements
elseif expression
 statements
else
 statements
end
```

```
a = 100;
%check the boolean condition
   if a == 10
      % if condition is true then print the following
      fprintf('Value of a is 10\n' );
   elseif( a == 20 )
      % if else if condition is true
      fprintf('Value of a is 20\n' );
   elseif a == 30
      % if else if condition is true
      fprintf('Value of a is 30\n' );
   else
      % if none of the conditions is true '
      fprintf('None of the values are matching\n');
   fprintf('Exact value of a is: %d\n', a );
   end
```

4.2 switch statement

```
switch <switch_expression>
case <case_expression>
<statements>
case <case_expression>
```

```
<statements>
...
...
otherwise
<statements>
end
```

```matlab
grade = 'B';
   switch(grade)
   case 'A'
      fprintf('Excellent!\n' );
   case 'B'
      fprintf('Well done\n' );
   case 'C'
      fprintf('Well done\n' );
   case 'D'
      fprintf('You passed\n' );
   case 'F'
      fprintf('Better try again\n' );
   otherwise
      fprintf('Invalid grade\n' );
   end
```

4.3 for loop

```
for variable = m:s:n

statement(s)

end
```

```matlab
x=[0, 2, 4, 6, 8 ];
n=length(x);
for i=1:1:n
y(i)=x(i)^2
end
```

4.4 while loop

```
While logical expression

Statement

end
```

```
x=2 ;

while x<20

x = 3*x-1

end
```

4.5 break statement

- break terminates the execution of for and while loops
- In nested loops, break terminates from the innermost loop only

```
y = 3;
for x = 1:10
  printf('%5d',x);
  if (x>y)
    break;
   end
end
```

5 Matrices

```
>> A= [ 1 2; 3 4] % defining a matrix
A =
 1   2
 3   4
>> B= [ 1 1; 1 1];

B =
 1   1
 1   1
```

5.1 Matrix Operations

```
>> A*B % the symbol \*" is matrix multiplication
ans =
  3   3
  7   7
>> A.*B % term by term multiplication
ans =
```

```
  1   2
  3   4
>> A^2 % taking square of A, which is A*A
ans =
  7   10
  15   22
>> A.^2 % taking term by term square
ans =
  1   4
  9   16
>> A.^4 % fourth power of each term
ans =
  1   16
  81   256
```

5.2 Special Matrices

Create the "identity matrix" of size 5

```
>> eye(5)
ans =

Diagonal Matrix

 1   0   0   0   0
 0   1   0   0   0
 0   0   1   0   0
 0   0   0   1   0
 0   0   0   0   1
```

Create a square matrix of ones

```
>> ones(4)
ans =

 1   1   1   1
 1   1   1   1
 1   1   1   1
 1   1   1   1
```

Create a 3x5 matrix of zeros

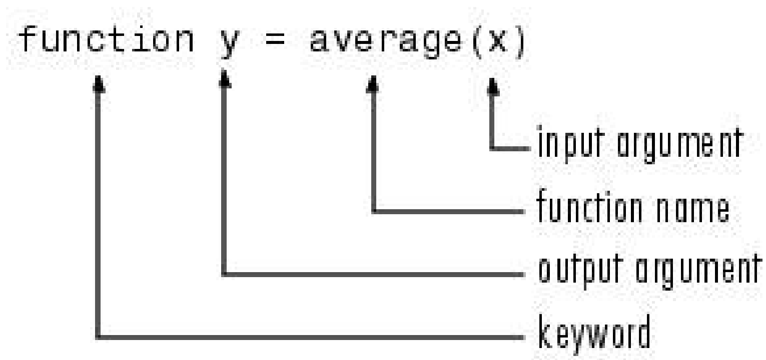
```
>> zeros(3,5)
ans =

 0   0   0   0   0
 0   0   0   0   0
 0   0   0   0   0
```

6 MATLAB Files

- MATLAB can execute a sequence of MATLAB statements stored on disk. Such files are called "M-files"
- These files must have the file extension of ".m"

7 User-defined Functions in MATLAB




```
function twosum(x,y)
  x+y
end
```

Save this function as twosum.m

```
>> twosum(2,2)
  ans =
     4
```

8 Lab01: Lab Tasks

1. Generate an all zero matrix, all ones matrix and random matrix using MATLAB.

2. Generate two random matrices A and B and find out the A*B and A.*B operation, describe the difference between these two operations.

3. Write a function that accepts temperature in degrees Fahrenheit (°F) and computes the corresponding value in degrees Celsius (°C). The relation between the two is

$$T^{\circ}C = \frac{5}{9}(T^{\circ}F - 32)$$

4. Calculate the area and perimeter of a circle and rectangle using MATLAB.

5. Equivalent Resistance The equivalent resistance R_{EQ} of three resistors in parallel is given by

$$R_{\mathrm{EQ}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

   Calculate the equivalent resistance R_{EQ} of the circuit assuming that $R_1 = 100\Omega$ $R_2 = 50\Omega$ and $R_3 = 40\Omega$.